

One camera network. One investigation workspace.

Sentinel • Gujarat Police Innovation Challenge

Camera registry, unified viewing and durable vehicle history.

MODEL 1 + MODEL 2

WORKING LOCAL IMPLEMENTATION

Engineering review • 14 September 2026

GPU analytics executes on organizer feeds. Public-sample recognition and alerts demonstrated; Indian accuracy remains unproven.

Connect the observation to an actionable record.

The designated-vehicle workflow



Provide ordered camera appearances with timestamps and locations. Preserve evidence and source provenance.

Registry and GIS are the mandatory foundation. Departmental VMS/storage continue operating independently.

From registry to vehicle history.

Four connected investigation pages

Registry & GIS

Manual/API/CSV onboarding, filters, metadata gaps and surveyed coverage.

Vehicle Trace

Registration search, time/source filters, ordered visits, map and CSV.

Watchlist

Representative or authorized records, updates, import and deactivation.

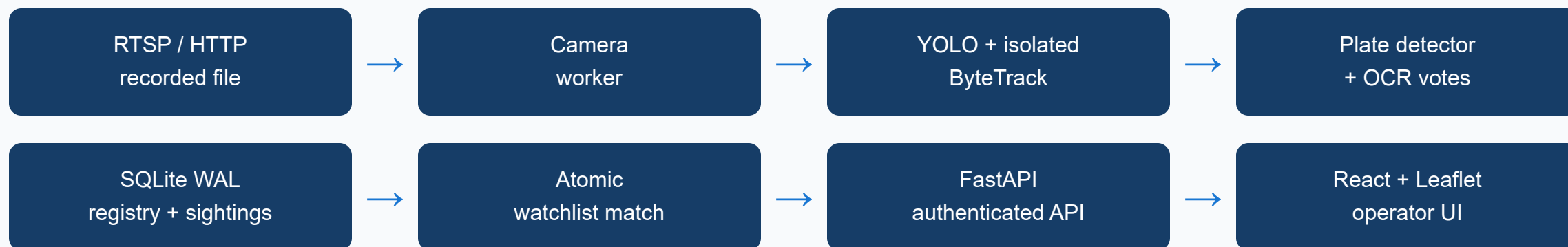
Alerts & Evidence

Automatic confirmed matches, crop inspection, acknowledgement and audit.

Map lines show camera observation order; they do not prove the road taken between observations.

Separate camera state. Share durable observations.

Implemented local architecture



A threaded single-service prototype. Regional workers, Kafka and PostGIS are the expansion design, not deployed components.

Keep uncertainty visible.

Controls built into the workflow

- Two independent qualified frames before plate confirmation.
- Separate trackers and sessions per camera.
- No person-to-vehicle class remapping.
- No silent local-video substitution for failed RTSP.
- Source type and timestamp basis retained.
- Uncertain reads do not raise confirmed-match alerts.
- Transaction-safe sightings and alerts.
- Unique evidence names and controlled file access.

Source camera resolution limits recognition. Scores are not calibrated identity probabilities.

Test behavior. Report what it proves.

Current evidence

Backend acceptance

62 passing checks: recognition, bounded OCR state, durable trace, roles, timing and responsive diagnostics.

Real browser checks

Camera/watchlist forms, synthetic history, surveyed coverage, downloads and acknowledgement.

Ingestion evidence

30 organizer cameras; cam06 RTSP/GPU run: 89 detection events in 120 seconds, zero confirmed plates.

Validation pending

Two complete Finnish scenes: 4/4 labelled appearances per scene; no false positives. Indian recognition and independent VMS integration remain unverified.

Browser and repository test records are synthetic fixtures in a separate database. They are not submission evidence of ANPR performance.

Process regionally. Investigate centrally.

Deployment path toward the statewide requirement

Scale design

- Regional ingestion and GPU workers.
- Durable event transport; PostgreSQL/PostGIS.
- Selected evidence in object storage.
- Measured capacity, backpressure and failover.

Security design

- Current: local restriction, sessions, roles and audit.
- Deployment: OIDC and department-scoped access.
- HTTPS, managed secrets and encrypted storage.
- Retention, backup and recovery exercises.

Illustrative: $80,000 \times 2 \text{ Mb/s} = 160 \text{ Gb/s}$ if all video is centralized. Actual sizing and costs require measurements and quotes.

Complete the evidence before claiming readiness.

Submission and next validation steps

1. Complete missing verified source metadata and geographic surveys.
2. Run a readable multi-camera vehicle journey and measure misses/false alerts.
3. Review completed public-source watchlist/alert and government detection videos with their reports.
4. Update this presentation and HLD with measured results.
5. Verify document/video link access and submit through the official portal.

LAST KNOWN: 15 SEPTEMBER 2026

EVENT: 22–23 SEPTEMBER

Saved official pages: 5–6 September. Recent site checks failed; verify dates in the portal.